

Unstructured Data.

No predefined format or schema - the data doesn't fit rows/columns at all.

Traits:

- ~80% of real-world data.
- Needs NLP/ML to process
- Hard to search directly.

Eg: Emails, PDFs, social media posts, satellite images.

Structured Data

Data that fits neatly into rows and columns with a fixed schema. Easiest to store, search and analyze.

Traits:

- Fixed fields / data types.
- Stored in RDBMS tables.
- Queryable via SQL.

Eg: Excel sheets, bank transactions, records, employee databases, etc..

Graph Based Data.

Focuses on relationships between entities rather than entities alone - stored as nodes (things) and edges (connections).

Traits:

- Nodes + edges + properties.
- Captures relationships well.
- Queried with graph DB's.

Eg: Social networks, recommendation engines, etc of data.

Facets of Data

raw facts and figures, structured or not.

Natural Language Data.

A special tricky case of unstructured data: human language itself, full of context, ambiguity, and grammar rules.

Traits:

- Needs NLP (tokenizing, parsing)
- Context-dependent meaning
- Basis for text mining, chatbots.

Eg: Reviews, transcripts, tweets, articles.



Machine Generated Data.

Created automatically by computers/sensors/apps with no human involvement - usually very well structured and huge in volume.

Traits:

- Very high volume and velocity.
- Consistent, predictable format.
- Growing fast with IoT.

Eg: Server logs, Sensor data, GPS data etc..

Audio, Image & Video.

Rich unstructured media that poses unique challenges for storage, indexing and analysis.

Traits:

- Large file sizes
- Needs specialized algorithms (CV, speech)
- Meta data helps searchability.

Eg: CCTV footage, MRI scans, podcasts.

Streaming Data.

Data generated continuously in real time, that must often be processed on the fly rather than stored first.

Traits:

- Continuous / time-stamped
- Needs Real time Processing
- High Velocity, perishable val.

Eg: Stock tickers, live sensor feeds etc.